Byeongkeun Kwon

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Kyungpook National University
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EDUCATION

2026.03–Present **Ph.D. Student in Data Science**
Graduate School of Data Science, Kyungpook National University, Korea
Advisor: Prof. Suhyeon Kim

2024.03–2026.02 **M.S. in Data Science**
Graduate School of Data Science, Kyungpook National University, Korea
Dissertation: *A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Index Estimation*
Advisor: Prof. Suhyeon Kim

2017.03–2024.02 **B.S. in Mathematics**
Pusan National University, Korea

2022.03–2024.02 **Interdisciplinary Major in Industrial Mathematics & Software**
Pusan National University, Korea

2023.03–2023.12 **Undergraduate Researcher**
Industrial Mathematics Center, Pusan National University, Korea

RESEARCH INTERESTS

- AI for mobility, industrial systems, and medical data
- Federated learning, privacy-preserving AI, and cybersecurity
- Risk estimation, anomaly detection, and decision-support modeling
- Graph machine learning and spatiotemporal representation learning

PUBLICATIONS

Notation. † equal contribution; * corresponding author.

SCI(E) / SSCI Journal Papers

[1] Eun-ji Lee†, **Byeongkeun Kwon**†, Suhyeon Kim*, Sung Hyun Noh*, "Predictive Modelling of the Need for Tracheostomy after Traumatic Cervical Spinal Cord Injury using Machine Learning," *Yonsei Medical Journal*, 2026. Co-first author; JCR Q1.

[2] **Byeongkeun Kwon**†, Dain Lee†, Hyunjun Ko, Hanbin Lee, Hyeonjun Hwang*, Suhyeon Kim*, "A Graph Neural Network Approach for Aquatic Biodiversity Prediction Leveraging Water System Interconnections," *Ecological Informatics*, May 2026. Co-first author; JCR Q1, 4.1%.

Scopus Papers

[1] **Byeongkeun Kwon**, Suhyeon Kim*, "Driver Behavior Anomaly Detection based on Federated Learning Considering Data Distribution Imbalance," *The Journal of Korean Institute of Communications and Information Sciences*, 50(3), 395–405, March 2025. First author; Best Paper of JKICS.

Working Papers

[1] **Byeongkeun Kwon**, Dain Lee, Suhyeon Kim*, "A Newtonian-informed Transformer Autoencoder for Fishing Vessel Navigation Risk Index Estimation." Under review at *Ocean Engineering*.

[2] **Byeongkeun Kwon**†, Dasom Seong†, Jiyun Park, Hyeonjun Hwang*, Suhyeon Kim*, "Water Quality Risk Index Estimation using Graph Neural Networks for Spatiotemporal Watershed Systems." Submitted to *Water Research*.

[3] **Byeongkeun Kwon**†, Dasom Seong†, Sung Hyun Noh*, Suhyeon Kim*, "SSIU-Net: Spine Screw Inverse U-Net for Anatomical Representation-guided Pedicle Screw Trajectory Planning." Manuscript in preparation. Target journal: *Medical Image Analysis*.

[4] **Byeongkeun Kwon**†, Sesun You†, Young Seop Son*, Suhyeon Kim*, "CAN-based Heavy-Duty Truck Weight Estimation: A Dual-phase Analysis of AI and Vehicle Dynamics." Manuscript in preparation. Target journal: *IEEE Transactions on Intelligent Vehicles*.

CONFERENCE PRESENTATIONS

International Conferences

1. **Byeongkeun Kwon**, Sesun You, Dain Lee, Dasom Seong, Hyun Jong Cho, Hyuntae Choe, Young Ok Lee, Young Seop Son, Suhyeon Kim, "CAN-based Heavy-Duty Truck Weight Estimation: A Dual-phase Analysis of AI and Vehicle Dynamics" *ICCAS 2026*, Sapporo, Japan, October 27–30, 2026.
2. Wonhee Kim, **Byeongkeun Kwon**, Suhyeon Kim, Young Ok Lee, Young Seop Son, "Multi-Rate Scheduling and Atomic-Step Decomposition for MLP-Mixer Inference on Bare-Metal Embedded Systems," *ICETA 2026*, Nha Trang, Vietnam, July 20–23, 2026.
3. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, "3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction," *MICCAI 2025 CLINICCAI*, Daejeon, Korea, September 23–27, 2025.
4. **Byeongkeun Kwon**, Eun-ji Lee, Suhyeon Kim, Sung Hyun Noh, "Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury," *ASIA SPINE & NCS 2025*, Busan, Korea, September 4–6, 2025.
5. Hyosung Kim, Taejun Kwon, **Byeongkeun Kwon**, Jieun Jang, Jihun Jang, "DRIVE: Drowsiness Recognition and Intelligent Vibration Evocation," *IEEE World Haptics Conference 2025*, Suwon, Korea, July 8–11, 2025.
6. **Byeongkeun Kwon**, Dain Lee, Suhyeon Kim, "Classical Mechanics Meet Deep Learning: Ship Hazardous Navigation Detection System," *IISE Annual Conference & Expo 2025*, Atlanta, USA, May 31–June 3, 2025.
7. Dain Lee, **Byeongkeun Kwon**, Hyunjun Ko, Suhyeon Kim, "A Graph Neural Network Approach for Aquatic Biodiversity Prediction Leveraging Water System Interconnections ," *IISE Annual Conference & Expo 2025*, Atlanta, USA, May 31–June 3, 2025.
8. **Byeongkeun Kwon**, Suhyeon Kim, "Baseball Match Prediction based on Graph Neural Network," *Deep Learn 2024: 11th International School on Deep Learning*, Porto, Portugal, July 15–19, 2024.

Domestic Conferences

1. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, "Graph Neural Network-based Water Quality Risk Index for Spatio-Temporal Watershed Systems," *Proceedings of KIIE 2026 Joint Conference*, Gyeongju, Korea, June 4–6, 2026.
2. **Byeongkeun Kwon**, Seseun You, Dain Lee, Dasom Seong, Suhyeon Kim, Hyung Jong Cho, Hyuntae Cho, Young Seop Son, Suhyeon Kim, "Lightweight MLP-Mixer for Real-time Weight-in-Motion with Embedded Deployment," *Proceedings of KSAE 2026 Spring Conference*, Jeju, Korea, May 20–23, 2026.
3. **Byeongkeun Kwon**, Dasom Seong, Suhyeon Kim, Sung Hyun Noh, "3D Generative Models for Vertebral Segmentation and Pedicle Screw Path Prediction," *The 44th Annual Spring Meeting of the Korean Neurosurgical Society*, Jeju, Korea, April 16–18, 2026.
4. **Byeongkeun Kwon**, Dain Lee, Dasom Seong, Suhyeon Kim, "Development of a Generative Adversarial Networks-based Fuzzing Test Technique for Vehicle Communication Security," *Proceedings of KSAE 2025 Fall Conference*, Busan, Korea, November 12–15, 2025.
5. **Byeongkeun Kwon**, Suhyeon Kim, "A Newtonian-informed Transformer AutoEncoder for Vessel Navigation Risk Estimation," *Proceedings of KIIE 2025 Fall Conference*, Daejeon, Korea, November 6–7, 2025.
6. **Byeongkeun Kwon**, Suhyeon Kim, "Mixer-integrated Graph Neural Networks for Accurate and Efficient Prediction of Crystalline Solid Properties," *Proceedings of KIIE 2025 Joint Conference*, Jeju, Korea, June 18–21, 2025. Poster.
7. **Byeongkeun Kwon**, Suhyeon Kim, "Privacy-preserving Federated Learning for Abnormal Driving Behavior Detection," *Proceedings of KICS Fall Conference 2024*, Gyeongju, Korea, November 20–22, 2024.
8. Eun-ji Lee, **Byeongkeun Kwon**, Suhyeon Kim, Sung Hyun Noh, "Development of Machine Learning Model Predicting the Need for Tracheostomy after Cervical Spinal Cord Injury," *NCS 2024*, Seoul, Korea, September 5–7, 2024.
9. **Byeongkeun Kwon**, Suhyeon Kim, "Baseball Match Prediction based on Graph Neural Network," *Proceedings of Korean Artificial Intelligence Association 2024 Conference*, Busan, Korea, August 16–17, 2024. Poster.

AWARDS AND HONORS

Grants

2026 Fall **KNU Ph.D. Fellow**, Kyungpook National University. Award amount: KRW 5,000,000 *(approximately USD 3.7K).*

2026 Spring **KNU Ph.D. Fellow**, Kyungpook National University. Award amount: KRW 5,000,000 *(approximately USD 3.3K).*

Awards

2025 **Minister of Science and ICT Award**, 2025 K-Data Science Conference & Hackathon, Hackathon Track.

2025 **Minister of Science and ICT Award**, 2025 ICT Challenge.

2025 **Minister of Environment Award**, 2025 Environmental Data Utilization & Analysis Competition, Analysis Track.

2024 **President Award of NIA**, 2024 K-Data Science Conference & Hackathon, Hackathon Track.

2024 **Chairperson Award of Korea National Park Service**, 2024 Environmental Data Utilization & Analysis Competition, Analysis Track.

2024 **Finalist Award**, 2024 Weather Big Data Contest, Fog Track.

2024 **Finalist Award**, 2024 Weather Big Data Contest, Power Track.

2023 **Finalist Award**, 2023 Weather Big Data Contest.

2023 **Encouragement Award**, 2023 Yeongnam Mathematical Society, Poster Presentation.

2023 **Excellence Award & Encouragement Award**, 2023 Mathematics May Competition, Departmental Contest.

PATENTS

1. Aquatic biodiversity prediction system and its method, 2025.

TEACHING ASSISTANTSHIPS

2026 Spring **Teaching Assistant**
Data Science Deep Learning (DASC0735), Graduate School of Data Science, Kyungpook National University

2026 Spring **Teaching Assistant**
Graph Data Analysis (DASC0767), Graduate School of Data Science, Kyungpook National University

ACADEMIC SERVICE

2025 Session Chair, *IISE Annual Conference & Expo 2025*, Atlanta, USA.

sectionResearch Experience and Projects

Ongoing Projects

2026.03–2031.02 **Development of Graph Deep Learning Core Technologies for Hypertrustworthy Dynamic Structural AI Countering Cybersecurity Threats**
National Research Foundation of Korea, Individual Basic Research / Excellent Young Researcher Type B — Researcher

2026.04–2028.03 **Advancement of an Adaptive AI-Based Industrial Control Protocol Security Testing Automation Platform and Expansion into Industrial Domains**
Industry-Academia-Research Collabo R&D Program, Commercialization R&D Program — Researcher

2025.03–2027.08 **BK21 FOUR Project: GLOW-AI Innovative Talent Development Education and Research Group**
BK21 FOUR — Researcher

2023.06–2026.05 **Sequential Data Analysis using Multi-view Graph Deep Representation Learning and Its Industrial Applications**
National Research Foundation of Korea, Creative Challenge Research Support Program — Researcher

2024.07–Present **Data-driven Intelligent Mobility Research Center**
IITP, ICT Research Center / ITRC Program — Researcher

2023.09–Present **Regional Demand Oriented Data Science Human Resource Development for Regional Industry Innovation**
Data Science Convergence Human Resource Development Program — Researcher

Completed Projects

2025.05–2025.12 **Generative AI Platform for Automated Analysis of Industrial Control Protocol Specifications and Automated Security Vulnerability Test-case Generation**
Industry-Academia-Research Collabo R&D Program, Preliminary Research Program — Researcher

2024.08– **GAN-based Automated Test-case Generation and Security Vulnerability Detection for Vehicle Communication Protocols and Infotainment File Formats**
Korean Ministry of SMEs and Startups, Startup Growth Technology Development Program — Researcher

REFERENCES

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